

IE 376, Fall 2025-2026

Homework #2

Due on November 27, 2025, Thursday by 5:00 pm

Question 1:

Read the attached paper `Enterprise resource planning (ERP) – A brief history` by Jacobs and Weston (especially Section 6. The future of ERP).

- a) Very briefly summarize the paper (at most one paragraph).
- b) How the generic ERP software packages can be tailored to specific market segments to simplify their implementations?
- c) Is it possible to make these systems `much more intelligent`, and how?

Question 2:

Consider the following jobs to be scheduled on a single machine. Assume that all jobs are available and the machine is idle at time 0.

Job	A	B	C	D	E
Processing Time	2	3	1	6	4
Due date	12	4	7	10	6

- a) What are the total tardiness, total flow time and maximum tardiness of the schedule that is obtained by using EDD rule?
- b) Start from schedule that you find in part (a) and obtain a schedule such that total tardiness is minimized using the greedy approach explained in class.
- c) Use the Apparent Tardiness Cost (ATC) rule discussed in class to obtain a schedule. Use the rule dynamically to dispatch one job at a time. What is the resulting total tardiness? Assume that the weights are all equal to 1 and use $k=2.0$.